

NDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Mining Engineering department

2. Subject Name: Ventilation and Climate Control L-T-P: 3-1-0
Credits:

3. Pre-requisites: None

Syllabus and reference books: Syllabus:

- Psychrometry; Human thermoregulation and heat stress indices.
- Estimation of mine heat load from various sources. Analytical and empirical rock to air heat transfer models.
- Design of mine air cooling systems.
- Basics of incompressible flow and mechanics of air flow distribution and flow control.
- Natural ventilation-estimation and thermodynamic understanding.
- Mine fans-basic theory of fan in creating air flow, types of fans, characteristic curves, operation in series and parallel, fan laws, fan drives, and installation. Fan selection methodology.
- Ventilation surveys, instruments, and ventilation planning of coal and metal mine ventilation systems.
- Auxiliary ventilation techniques, blast fume clearance and re-entry time estimation
- Ventilation economics and ventilation optimisation through controlled recirculation and ventilation on demand (VoD) techniques.

Reference Books:

1. Misra G. B. (1986). Mine Environment and Ventilation, OUP.
2. Banerjee S. P. (2003). Mine Ventilation, Lovely Prakashan, Dhanbad.
3. McPherson M. J. (1993). Subsurface Ventilation and Environmental Engineering, Chapman & Hall
(<https://link.springer.com/content/pdf/10.1007%2F978-94-011-1550-6.pdf>) or
https://dxi97tvbmhbca.cloudfront.net/upload/user/image/English_V-Subsurface_Ventilation_Engineering-SRKS_Mine_Ventilation_Services20200220222420293.pdf
4. <https://sites.google.com/site/mineventilationiitkgp/>

4. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Psychrometry; Human thermoregulation and heat stress indices.	5
2.	Estimation of mine heat load from various sources. Analytical and empirical rock to air heat transfer models.	5
3.	Design of mine air cooling systems.	3
4.	Basics of incompressible flow and mechanics of air flow distribution and flow control	7
5.	Natural ventilation-estimation and thermodynamic understanding.	3
6.	Mine fans-basic theory of fan in creating air flow, types of fans, characteristic curves, operation in series and parallel, fan laws, fan drives, and installation. Fan selection methodology.	7
7.	Ventilation surveys, instruments, and ventilation planning of coal and metal mine ventilation systems.	5
8.	Auxiliary ventilation techniques, blast fume clearance and re-entry time estimation.	3
9.	Ventilation economics and ventilation optimisation through controlled recirculation and ventilation on demand (VoD) techniques.	5
Total number of hours		43